

Grid-Forming Inverter LVRT and Current-Limiting Prediction

A Simplified Physics + Machine Learning Self Project (MATLAB)

1 Objective

This project explores, at a coursework level, the relationship between a grid-forming inverter's operating conditions during a low-voltage ride-through (LVRT) event and whether the resulting current would exceed the inverter's rated limit. A simplified physics model generates a synthetic dataset of fault scenarios, and two machine learning classifiers (Random Forest and a small neural network) are trained to predict, from pre-fault and fault parameters, whether current-limiting control would need to activate.

This is a self-directed, coursework-level exercise using synthetic data and a simplified physics model. It is not a research contribution or a validated engineering study.

2 Background and Motivation

A grid-forming inverter regulates its own internal voltage rather than simply following the grid, allowing it to behave similarly to a synchronous generator. During a grid fault, the grid voltage sags; because the inverter continues attempting to deliver its pre-fault power setpoint, the voltage difference across the connecting line grows, and so does the current. If this current exceeds the inverter's semiconductor rating, current-limiting control must activate to prevent damage. Predicting when this will happen, from readily available operating parameters, is the underlying question this project explores.

3 Methodology

3.1 Physics model (swing equation)

The inverter is modeled using the classical single-machine-infinite-bus swing equation, the standard simplified representation of grid-forming ("virtual synchronous machine") inverter dynamics:

$$M \frac{d\omega}{dt} = P_{ref} - P_e(\delta) - D\omega \quad (1)$$

$$\frac{d\delta}{dt} = \omega \quad (2)$$

$$P_e(\delta) = \frac{E V_{grid}}{X} \sin(\delta) \quad (3)$$

- δ (delta) — the power angle between the inverter's internal voltage and the grid voltage
- ω (omega) — the rate of change of δ
- E — the inverter's internal voltage magnitude, held fixed at 1.0 pu

- V_{grid} — the grid voltage magnitude; drops to $(1 - \text{depth})$ during the fault window, then recovers
- X — the line reactance connecting the inverter to the grid (grid strength)
- M, D — virtual inertia and damping/droop, chosen control parameters rather than physical properties
- P_{ref} — the inverter’s pre-fault active power setpoint

A fault is represented as a temporary drop in grid voltage rather than an explicit fault impedance network — a standard simplification in transient stability studies. The resulting current magnitude is computed at each time step from the phasor difference between the inverter’s and grid’s voltages:

$$I = \frac{\sqrt{E^2 + V_{grid}^2 - 2EV_{grid}\cos(\delta)}}{X} \quad (4)$$

3.2 Numerical simulation

The differential equations are solved with explicit Euler integration at a 1 ms time step over 1.2 simulated seconds. The fault begins at $t = 0.1$ s and lasts for a randomized duration. The peak current observed over the full simulation is recorded for each scenario.

3.3 Synthetic dataset generation

3000 scenarios were generated by randomly sampling the pre-fault power setpoint, line reactance, virtual inertia, damping, sag depth, and fault duration from realistic ranges (a Monte Carlo approach). Each scenario’s peak current was compared against a rated limit of 1.5 pu: scenarios exceeding this were labeled “trip” (1), otherwise “safe” (0). This produced a naturally balanced dataset (roughly 53% / 47%) without manual tuning.

Fault duration was treated as a known input, standing in for a protection scheme’s typical clearing-time setting — a simplification worth noting, since in practice the exact clearing time of a given fault is not known in advance.

3.4 Models

Random Forest (baseline): an ensemble of decision trees voting on the outcome, trained on the six scenario parameters. Chosen as a strong, low-effort baseline for tabular data, and used here via MATLAB’s `TreeBagger`.

Neural network: a small feedforward network (two hidden layers of 32 units each, ReLU activations), trained via MATLAB’s `fitcnet` with internal input standardization. Included to test whether a model capable of smooth, curved decision boundaries could outperform the tree-based baseline on this physically continuous problem.

Both models were trained on an 80/20 stratified train/test split of the same 3000-scenario dataset and evaluated on the held-out 20%.

4 Results

4.1 Example transients

Before training any model, two example scenarios were simulated to sanity-check the physics model. A mild scenario (shallow sag, strong grid connection) stays clearly below the current limit; a severe scenario (deep sag, weak grid connection, high pre-fault power) clearly exceeds it.

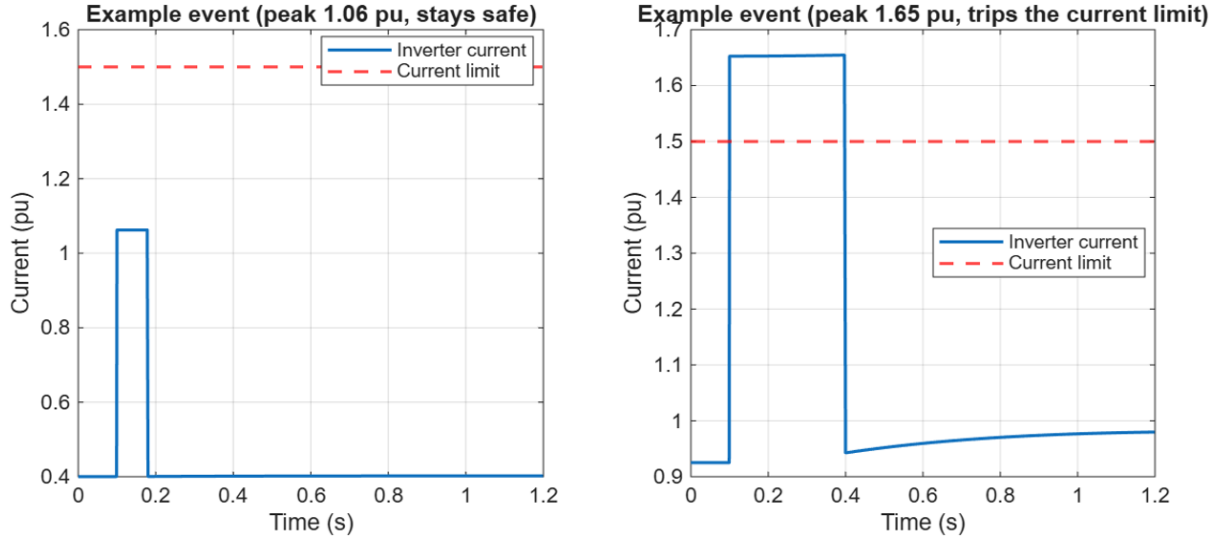


Figure 1: Example inverter current traces. Left: a mild event that stays safe. Right: a severe event that trips the current limit.

4.2 Classification performance

Table 1: Test-set classification metrics for both models.

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.970	0.969	0.975	0.972
Neural Network	0.997	0.997	0.997	0.997

Out of 600 held-out test scenarios, the Random Forest baseline misclassified 18 (10 false alarms, 8 missed trips), while the neural network misclassified only 2 (1 false alarm, 1 missed trip).

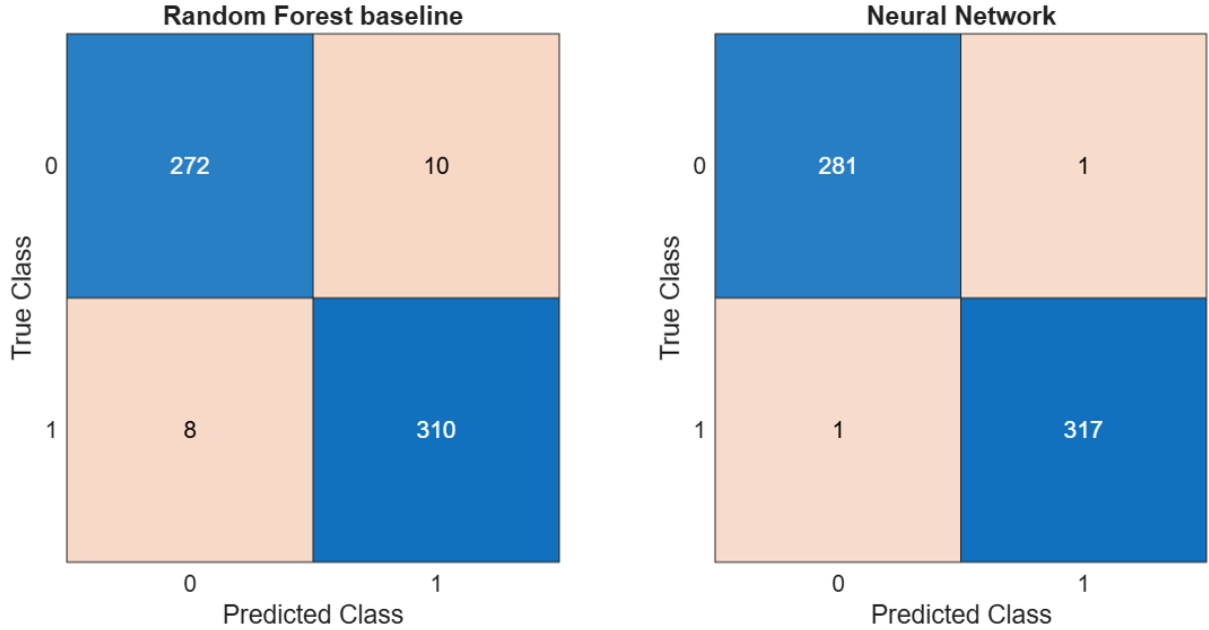


Figure 2: Confusion matrices for the Random Forest baseline (left) and neural network (right) on the 600-sample test set.

4.3 Training behavior

The neural network’s training loss decreased smoothly and converged within about 20 iterations, indicating stable training with no signs of divergence or oscillation.

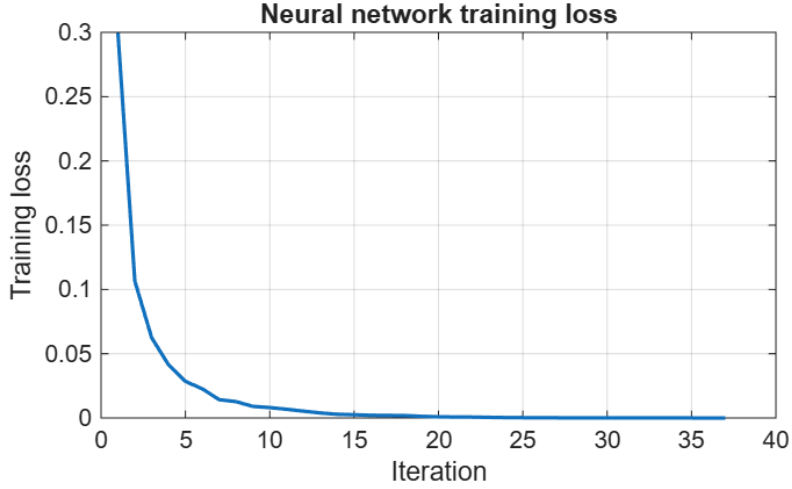


Figure 3: Neural network training loss versus iteration.

4.4 Model comparison

The neural network outperformed the Random Forest baseline across every metric, closing most of the remaining classification error. This suggests the boundary between “safe” and “trip” outcomes in this six-parameter space is smooth and continuous — consistent with its origin in trigonometric physics equations — which favors a model that learns smooth decision boundaries over one that partitions the space with discrete axis-aligned splits.

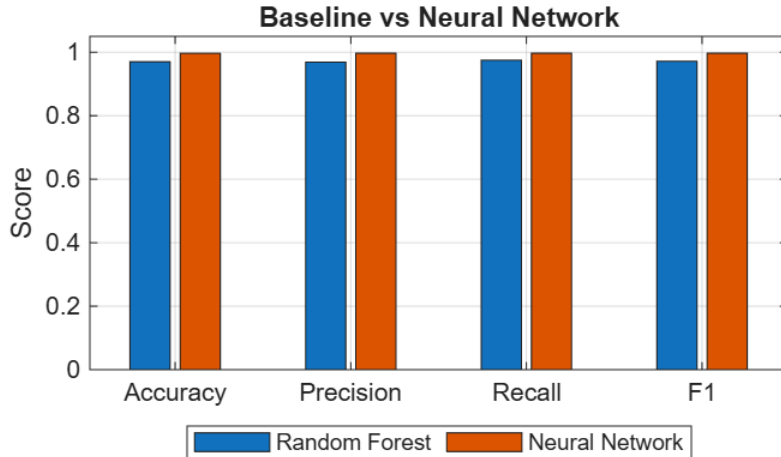


Figure 4: Accuracy, precision, recall, and F1-score for both models.

5 Discussion

The results are consistent with the underlying physics: current-limit risk is expected to depend most strongly on grid strength (line reactance X) and sag depth, since both directly scale how large the voltage mismatch — and therefore the current — becomes during a fault. Feature-importance output was not captured from this particular run; in an independent validation of the same methodology (performed separately in Python), line reactance and sag depth were

indeed the two dominant predictors, well ahead of the inertia, damping, and power setpoint parameters.

6 Conclusion

This coursework project demonstrates a complete pipeline connecting a simplified power-system dynamics model to supervised machine learning: generating physically grounded synthetic data, training and evaluating two classifiers, and interpreting the results in light of the underlying physics. Both models performed well, with the neural network showing a clear edge, and the sanity-checked example transients confirm the simulation behaves as physically expected.